Karlsruher Institut für Technologie

SPS Wake Model Vertical Growth Rate

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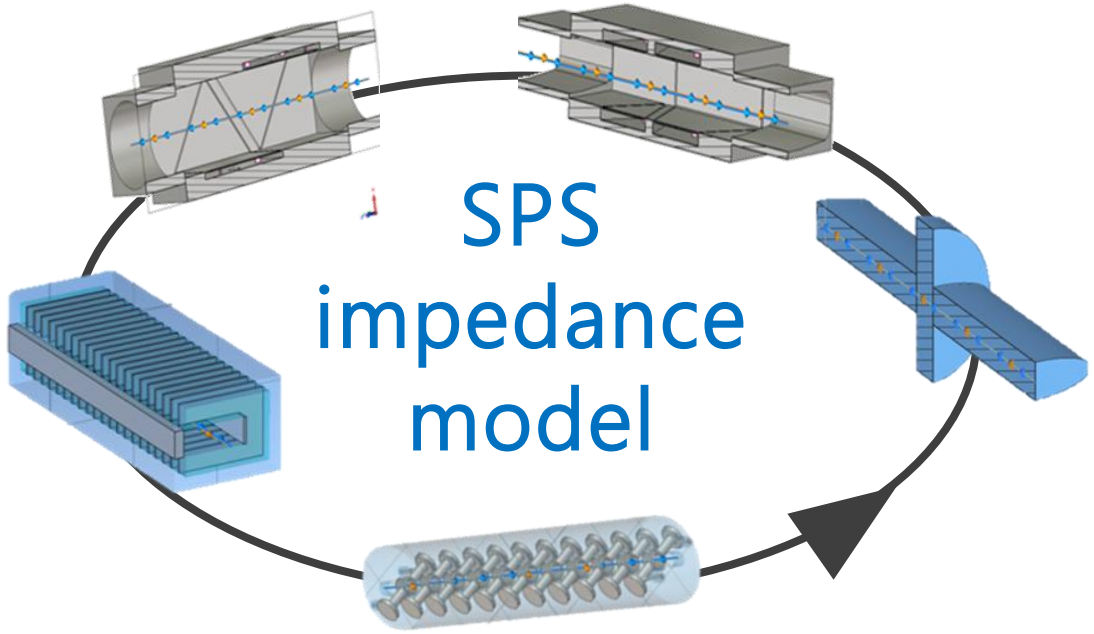
25.06.2026

Outline

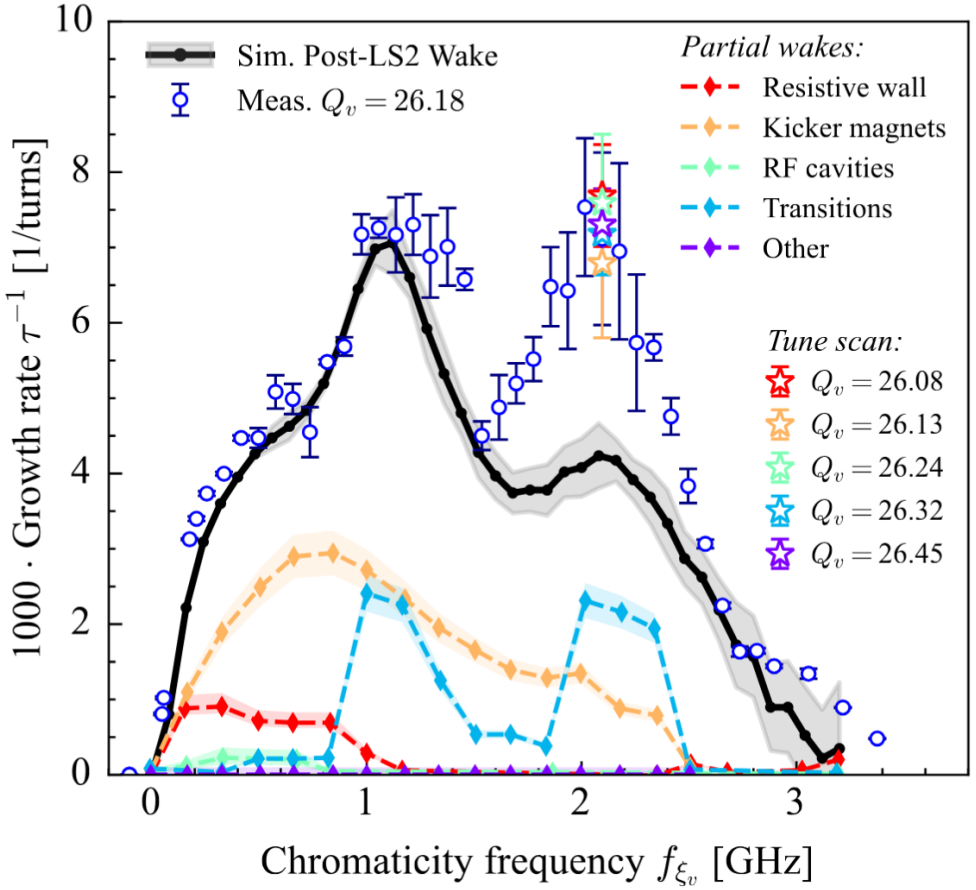
- Dark Impedance
- Measurements
- Simulations
- Wake Function Extrapolation

SPS Impedance Model

- Has been successfully benchmarked both for growth rates and tune shifts
- New approach of measurements so higher frequencies above 1,5 GHz have been measured for the first time
- For the SPS the Impedance model is sufficient
- Above cutoff will be important for FCC so we are interested in understanding that frequencies



Dark Impedance

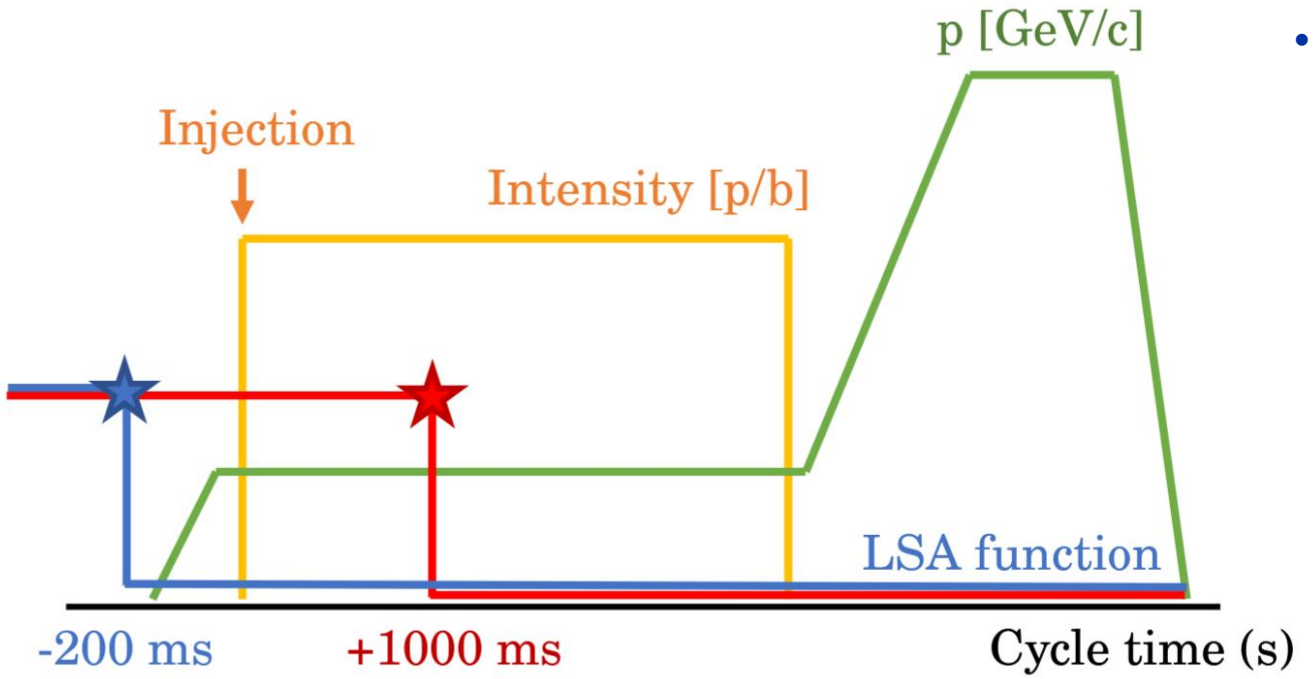


- „Dark Impedance“ above 2 GHz
- Transitions contribute to impedance at that frequency
- Look for „Dark Impedance“ in the simulation of the transitions

E. de la Fuente et al. (2025). https://doi.org/10.18429/jacow-ipac2025-wepm068

Measurements

Measurement Methodology

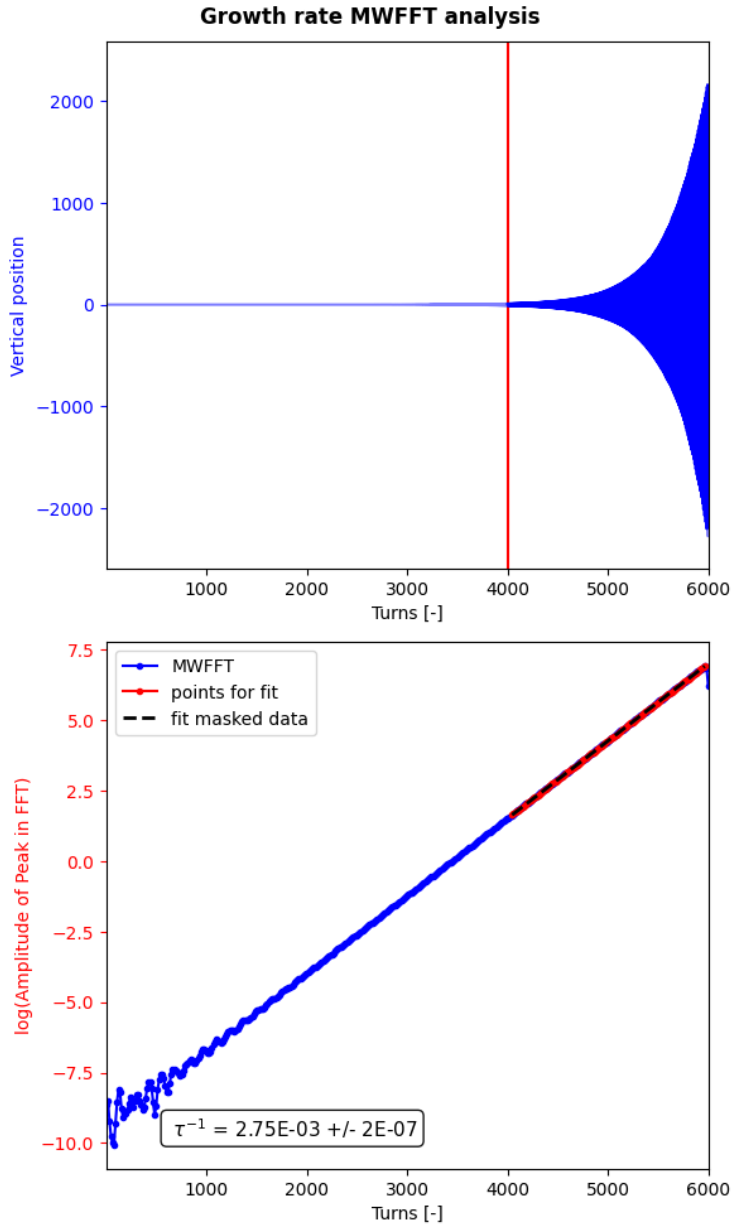


- Chromaticity change after injection for measuring higher frequencies

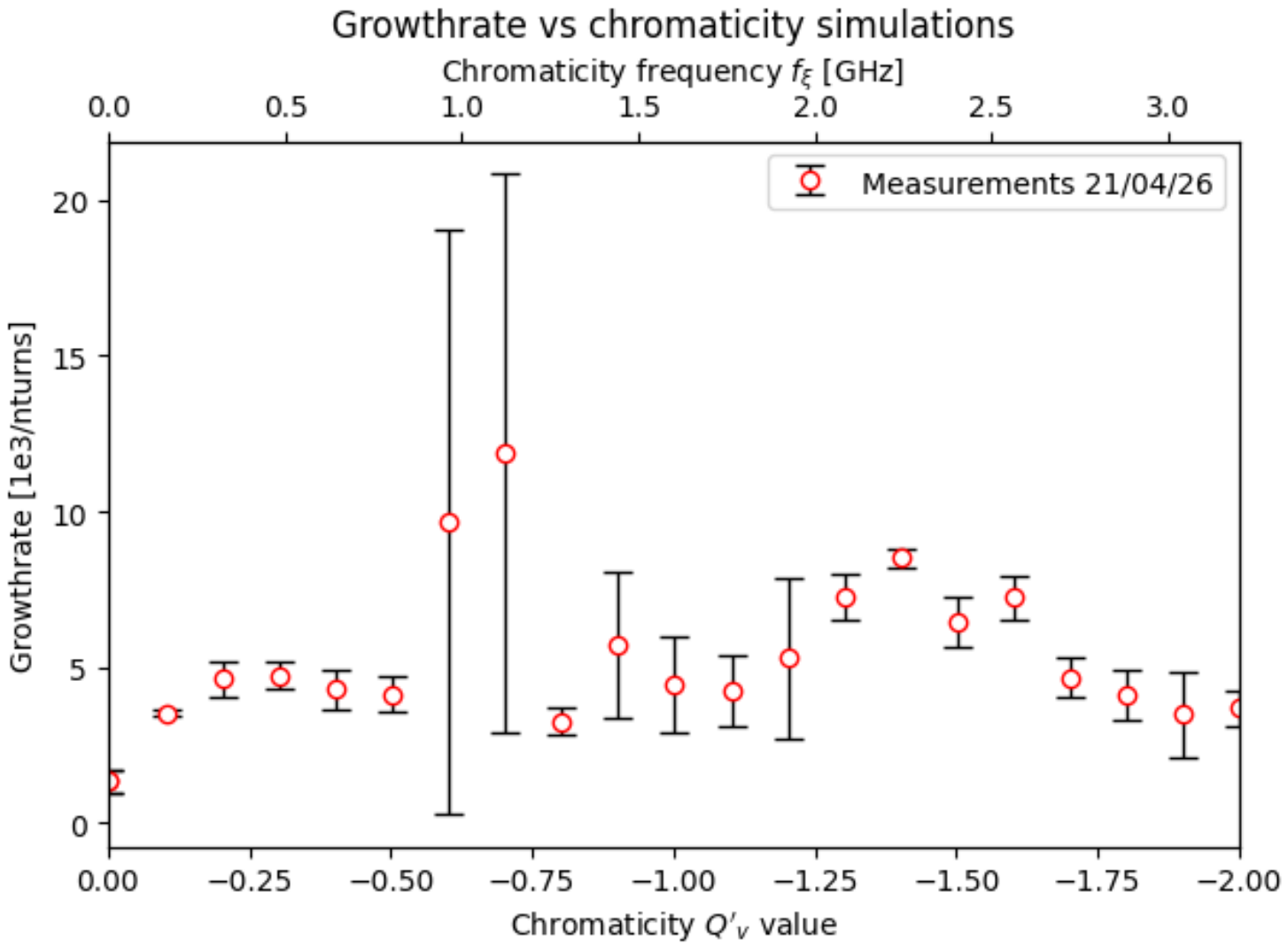
E. de la Fuente et al. (2023). https://doi.org/10.18429/JACoW-IPAC2023-WEPL155

Signal Processing

- Average bunch position during cycle
- Exponential fit to average bunch position

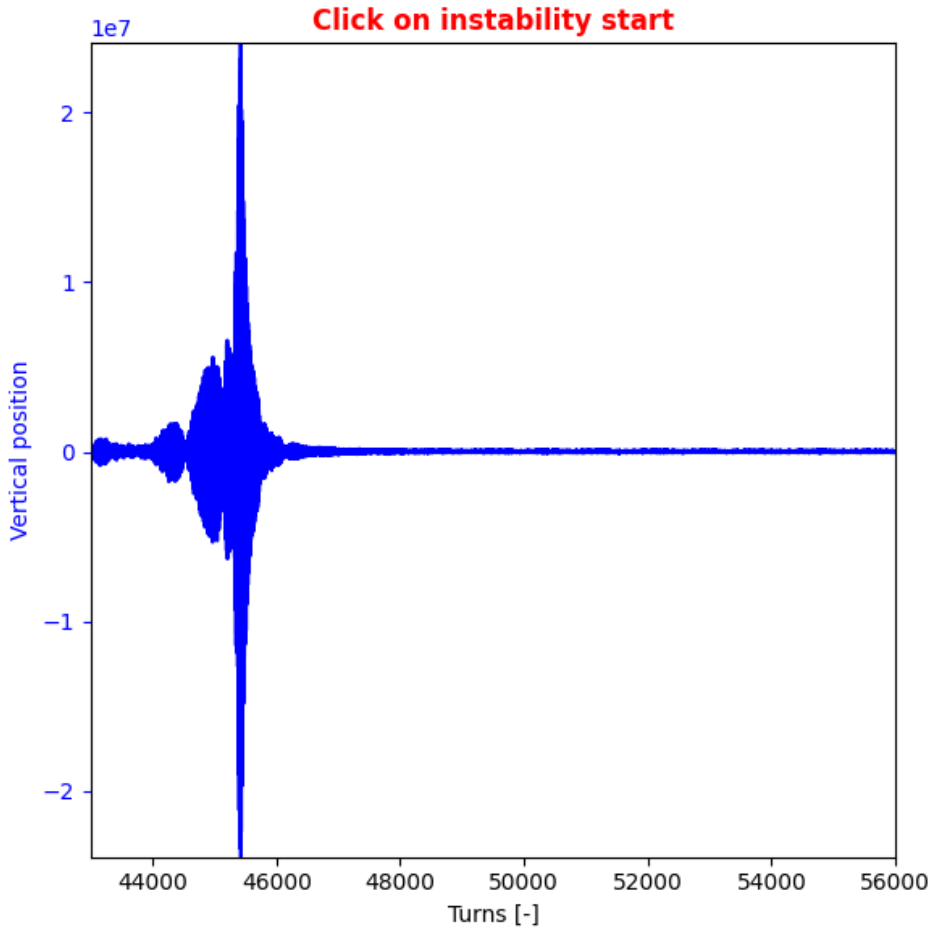


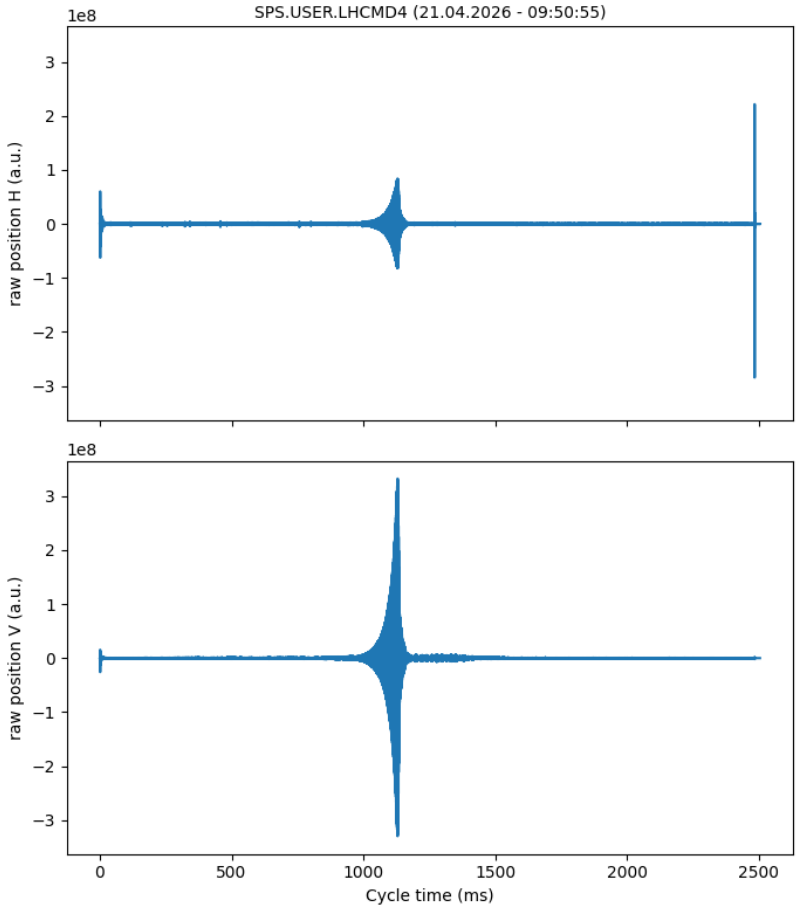
First Measurement



- High uncertainties

Horizontal Coupling



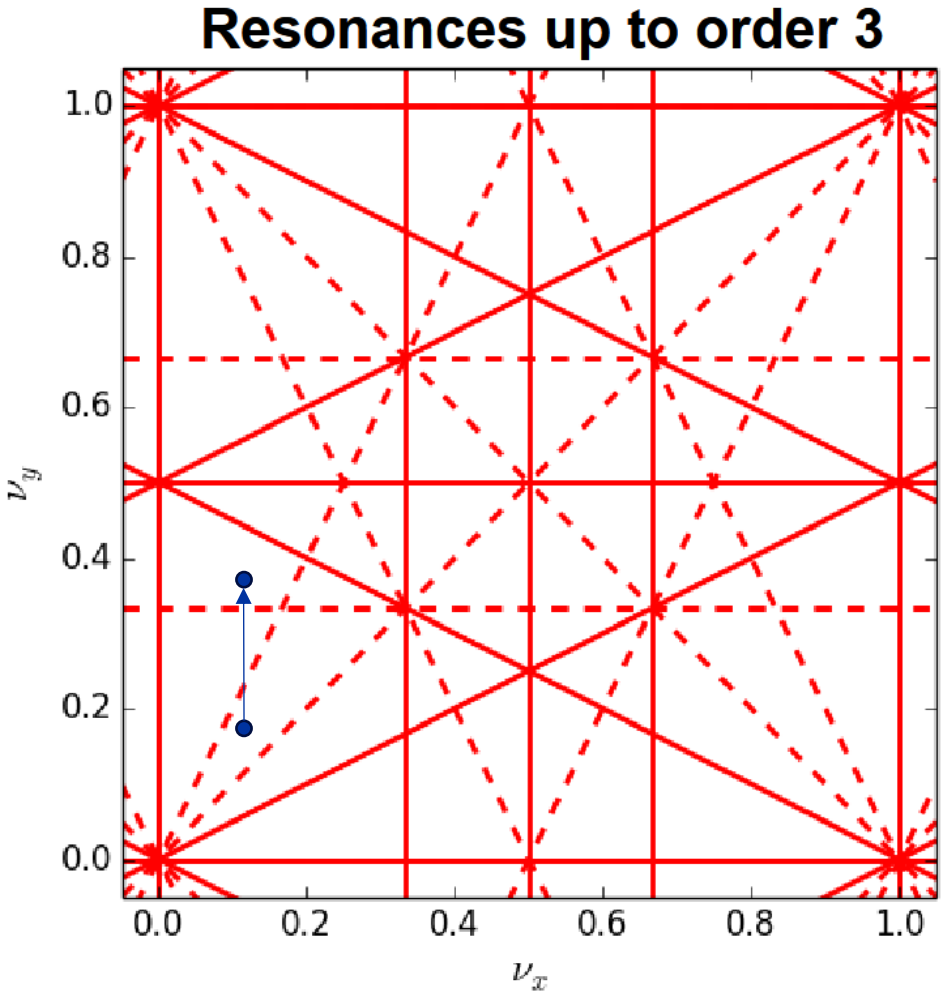


- No clean exponential growth
- Kick in vertical leads to growth in horizontal

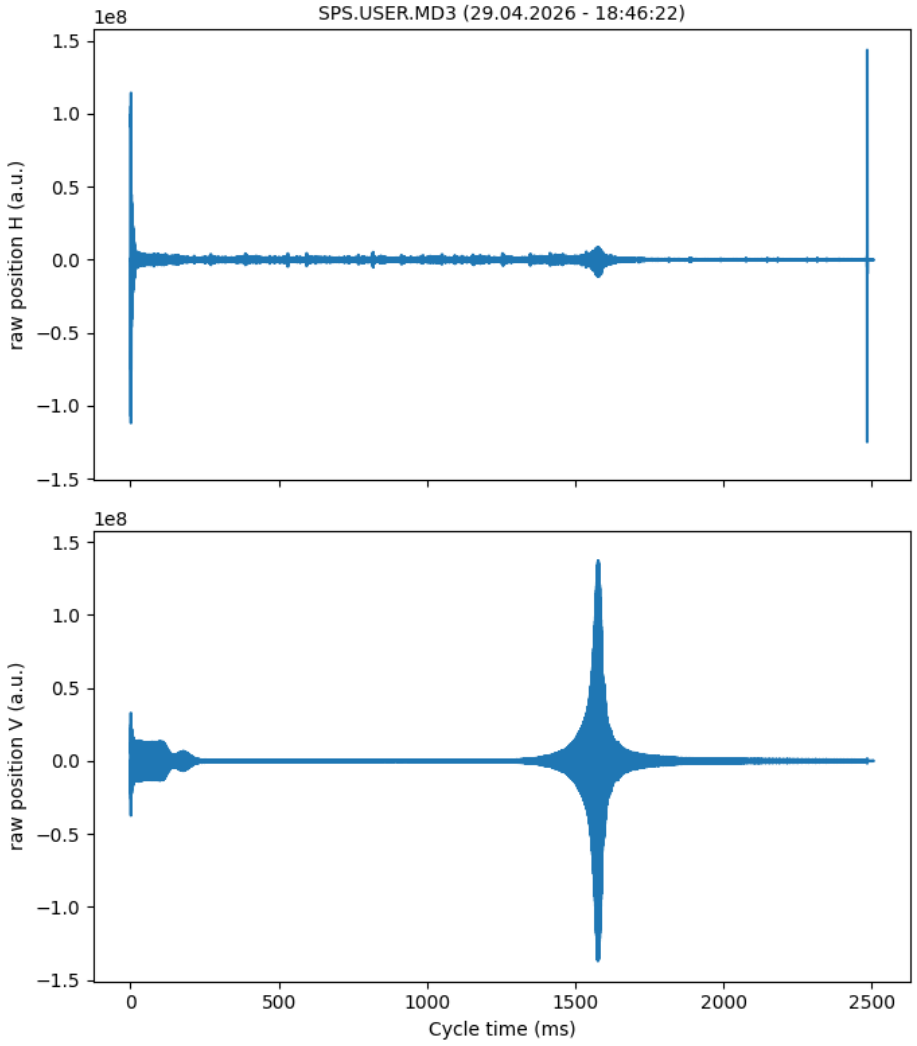
=> strong coupling between vertical and horizontal

Separation of Tunes

- Different tune chosen to separate tunes
  - Measurement 1:
    - Qx=26.13
    - Qy=26.18
  - Measurement 2:
    - Qx=26.1396
    - Qy=26.3711

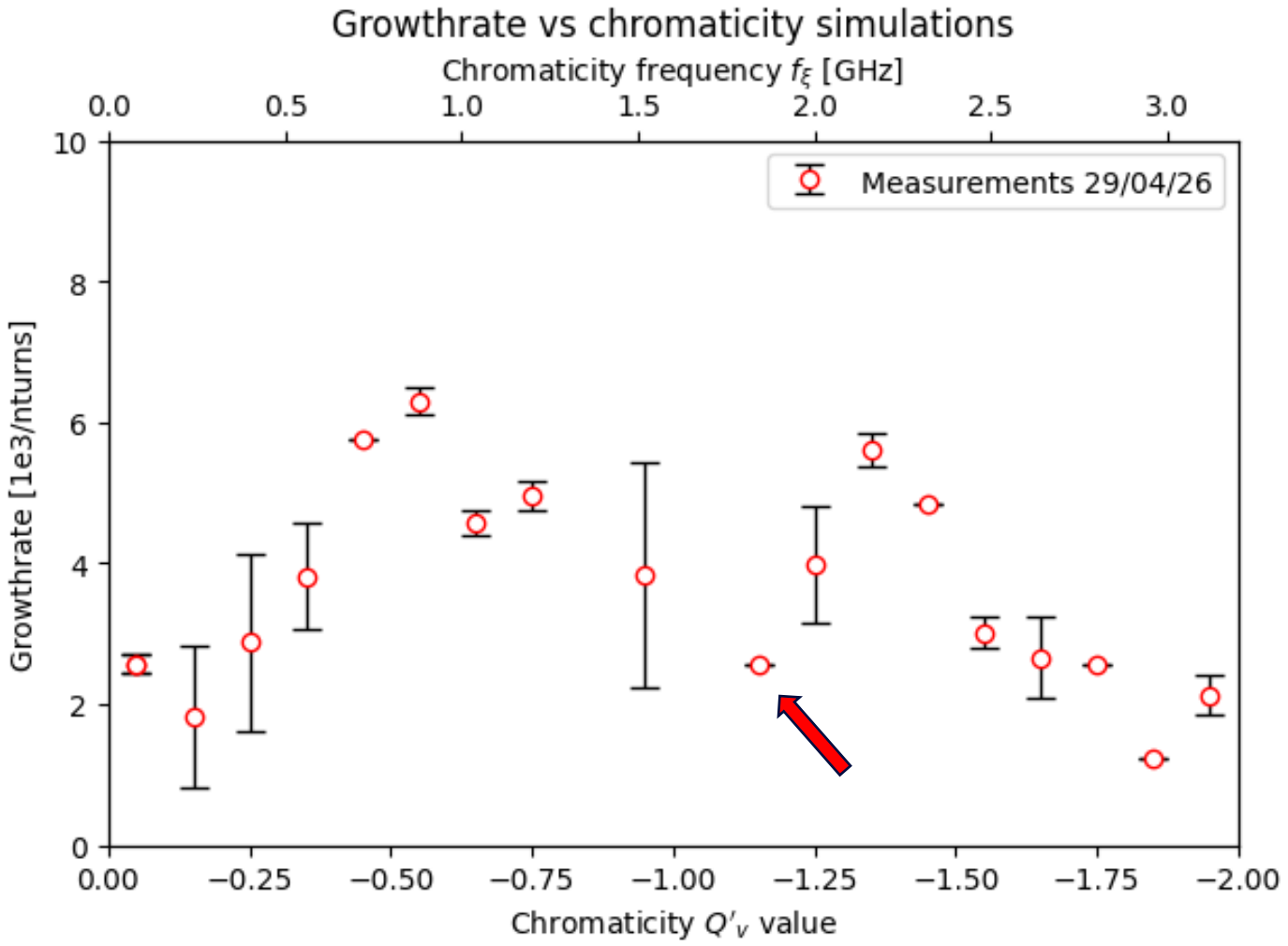


Coupling between H and V

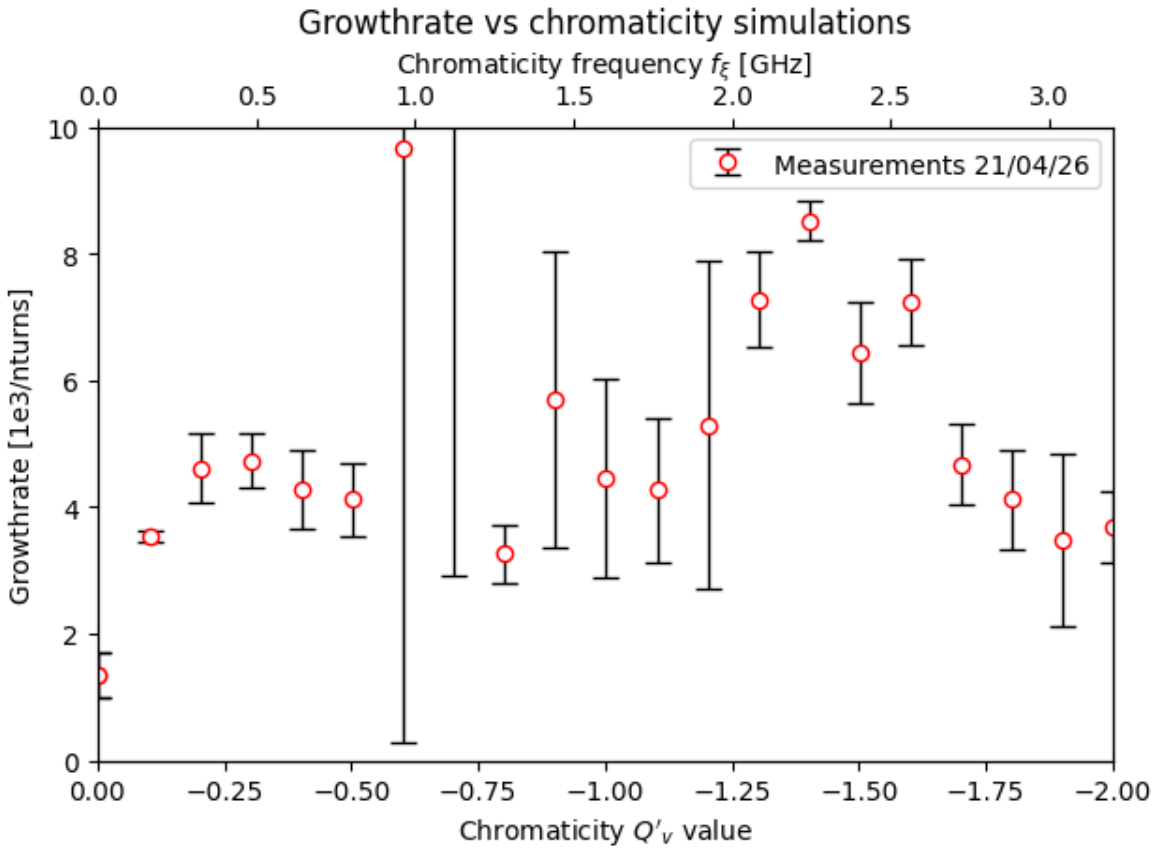


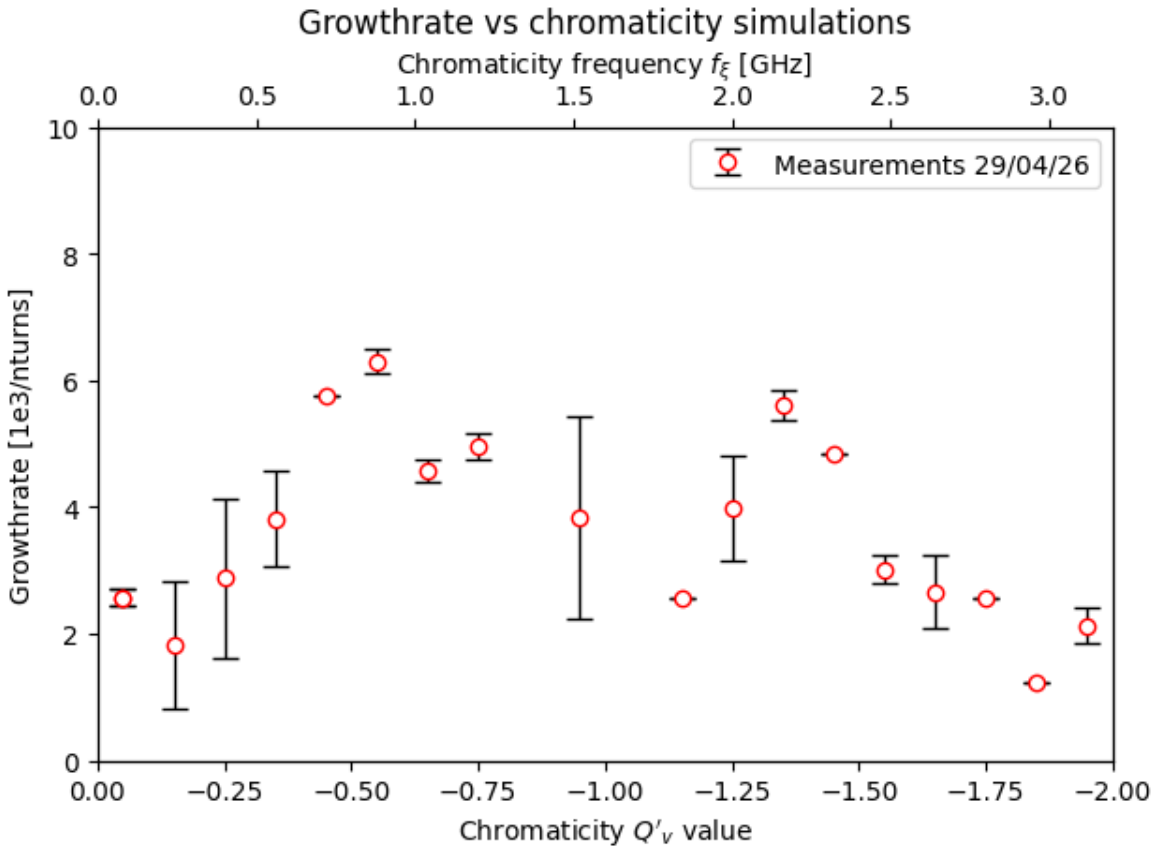
- Kick in vertical layer leads to less growth in horizontal layer

Second Measurement



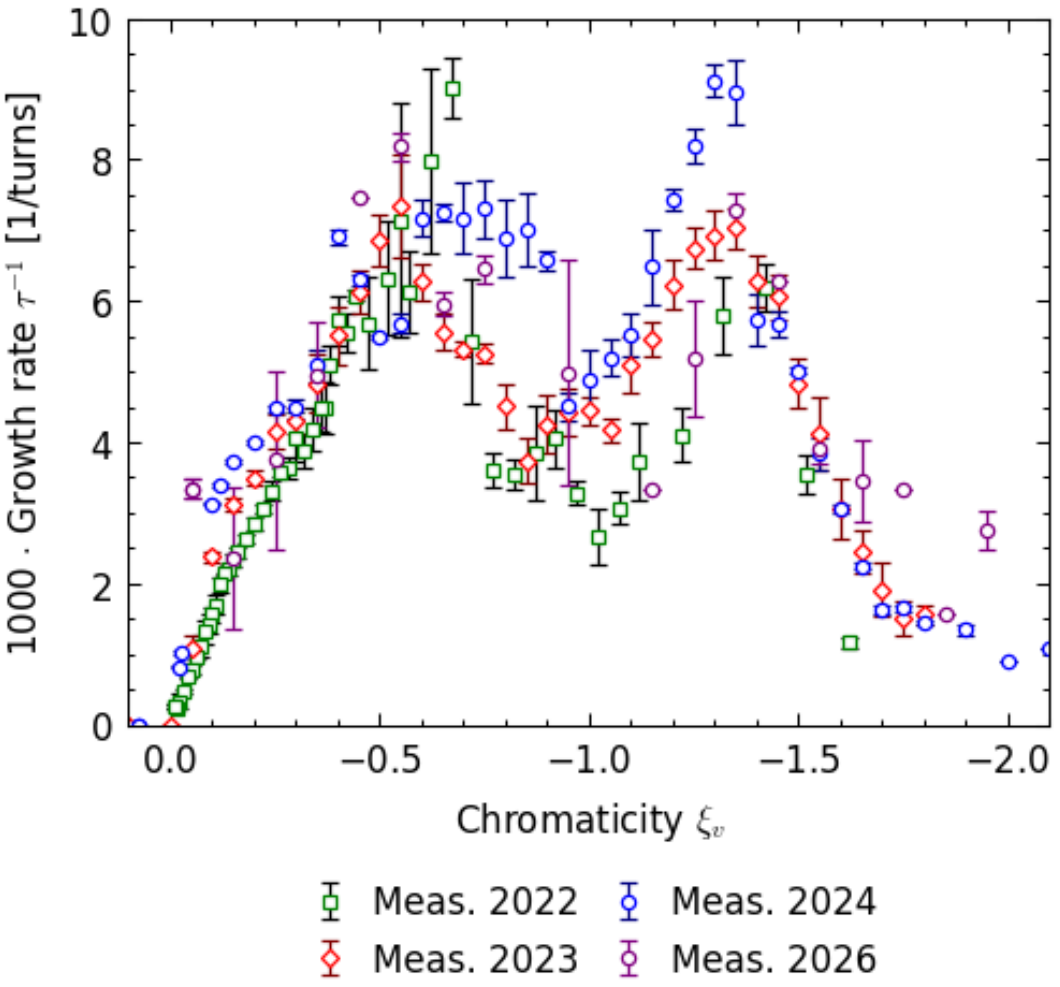
- Few usable data points





- Second peak still appears

Measurements of the Last Years



- Measurements done in the previous years under different conditions
- Second measurement of this year included in range of old measurements
- Normalized to same intensity in low frequency range

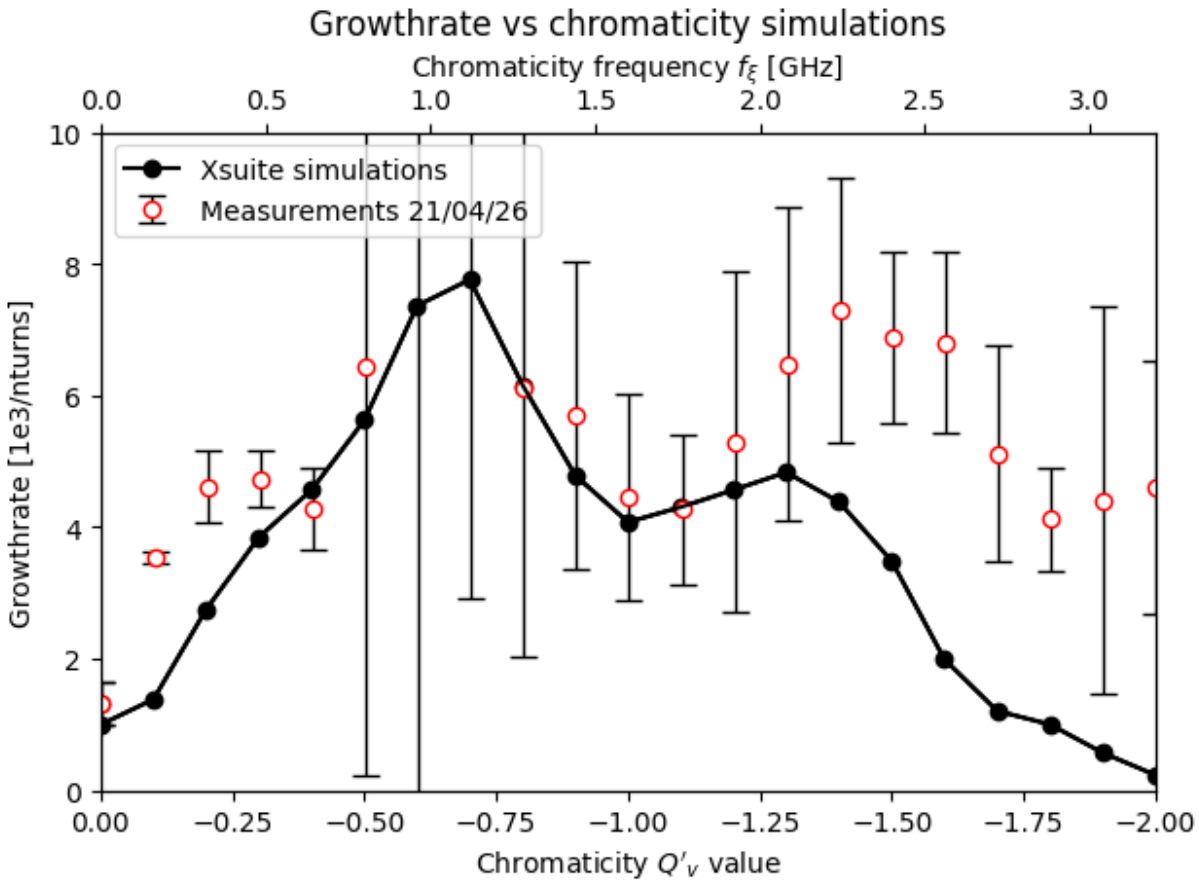
- Second peak varies under different conditions but is there in any case

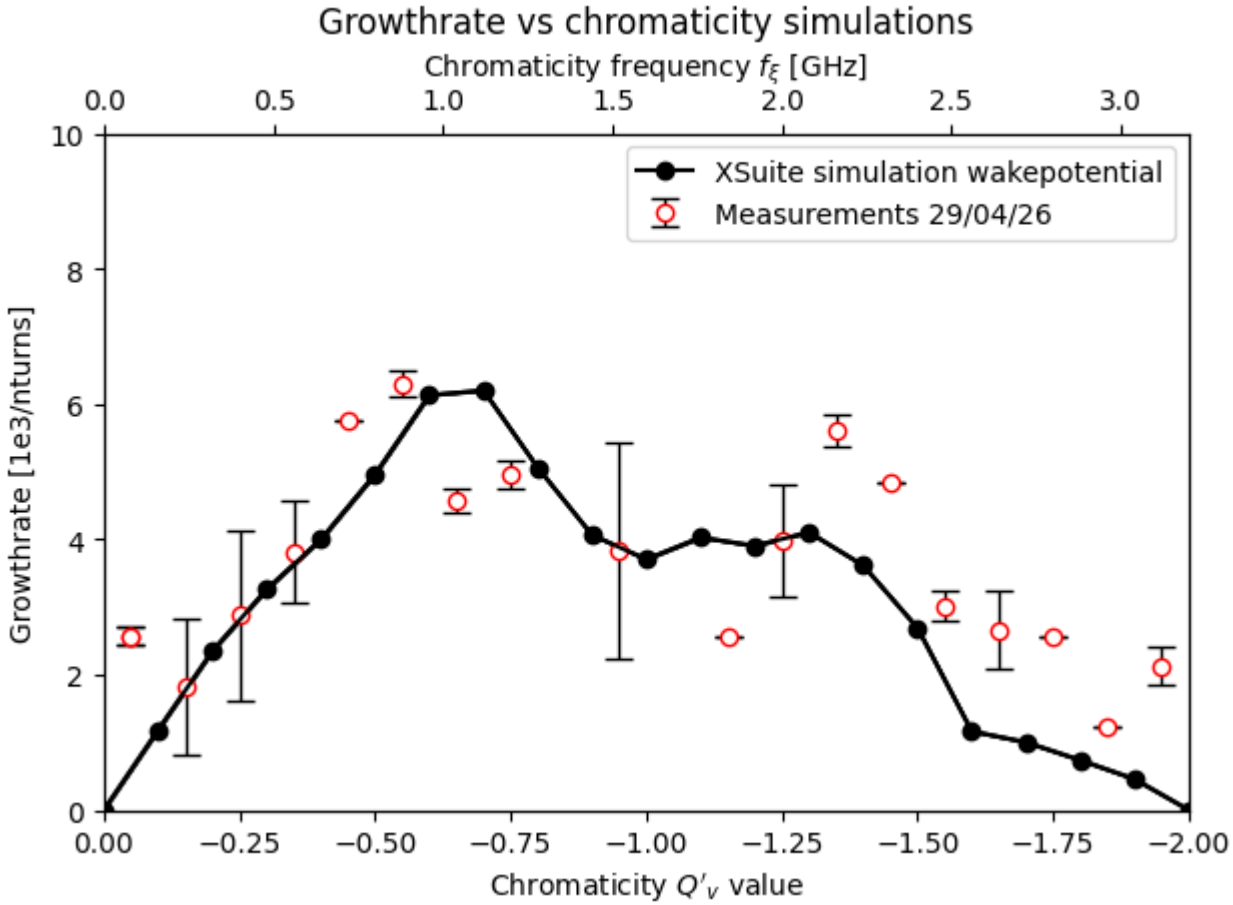
Simulation

Simulation Parameters

	Measurement 1	Measurement 2
Chromaticity (H / V)	0.1 / -2 … 0	0.1 / -2 … 0
Tune (H / V)	26.13 / 26.18	26.1396 / 26.3711
Intensity	3.6e10	2.6e10
Bunch length [ns]	3.2	2.8
Emittance	1.5e-6	1.5e-6
Turns	6000	6000
Turns of wake	5	1
Slices	300	300, integrated
Macroparticles	1e6	1e6
Nonlinear Chroma	X: 612.213*2, -90760.404*6 Y: -55.902*2, +55572.440*6	X: 612.213*2, -90760.404*6 Y: -55.902*2, +55572.440*6

Simulation Fit

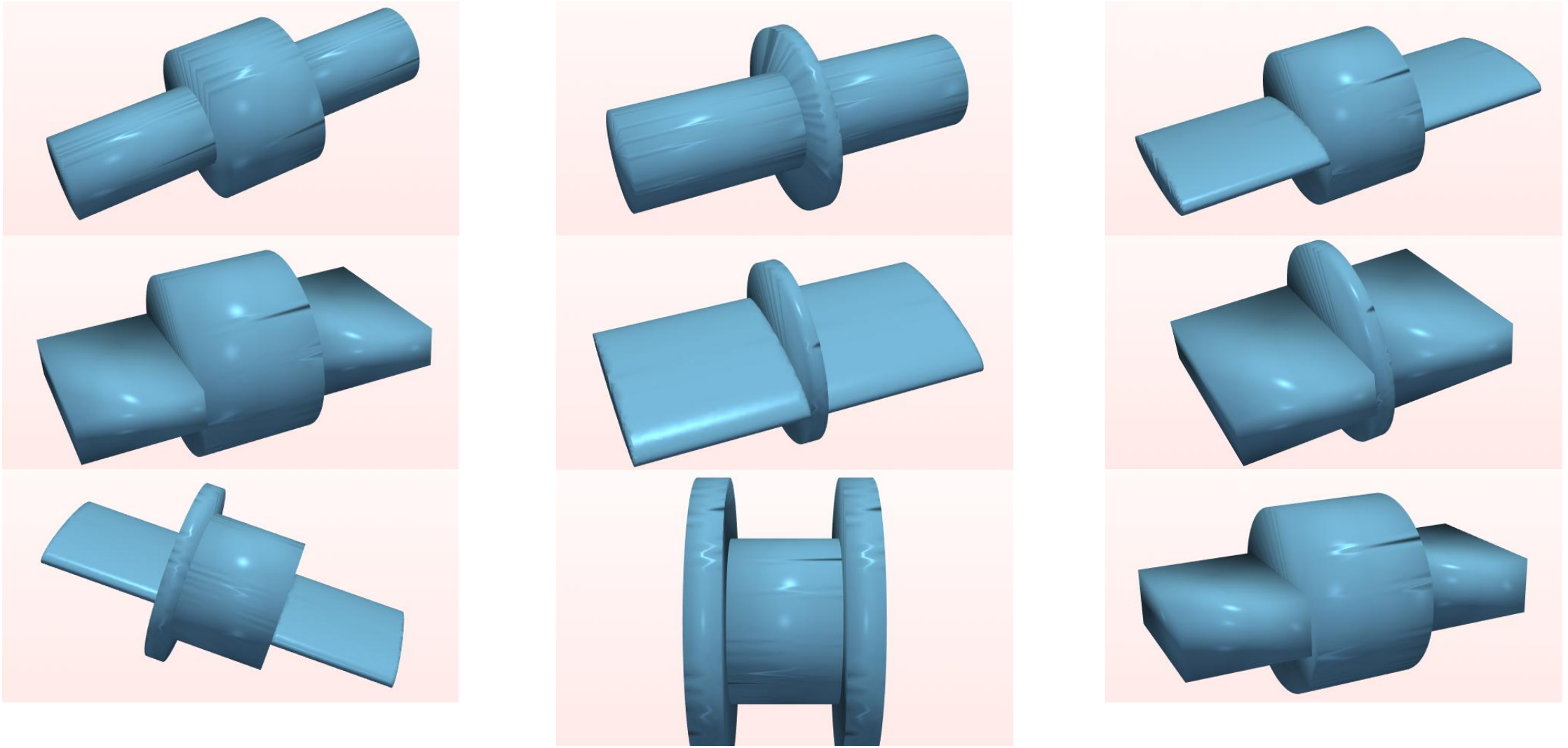




- „Dark Impedance“ above 2 GHz appears again

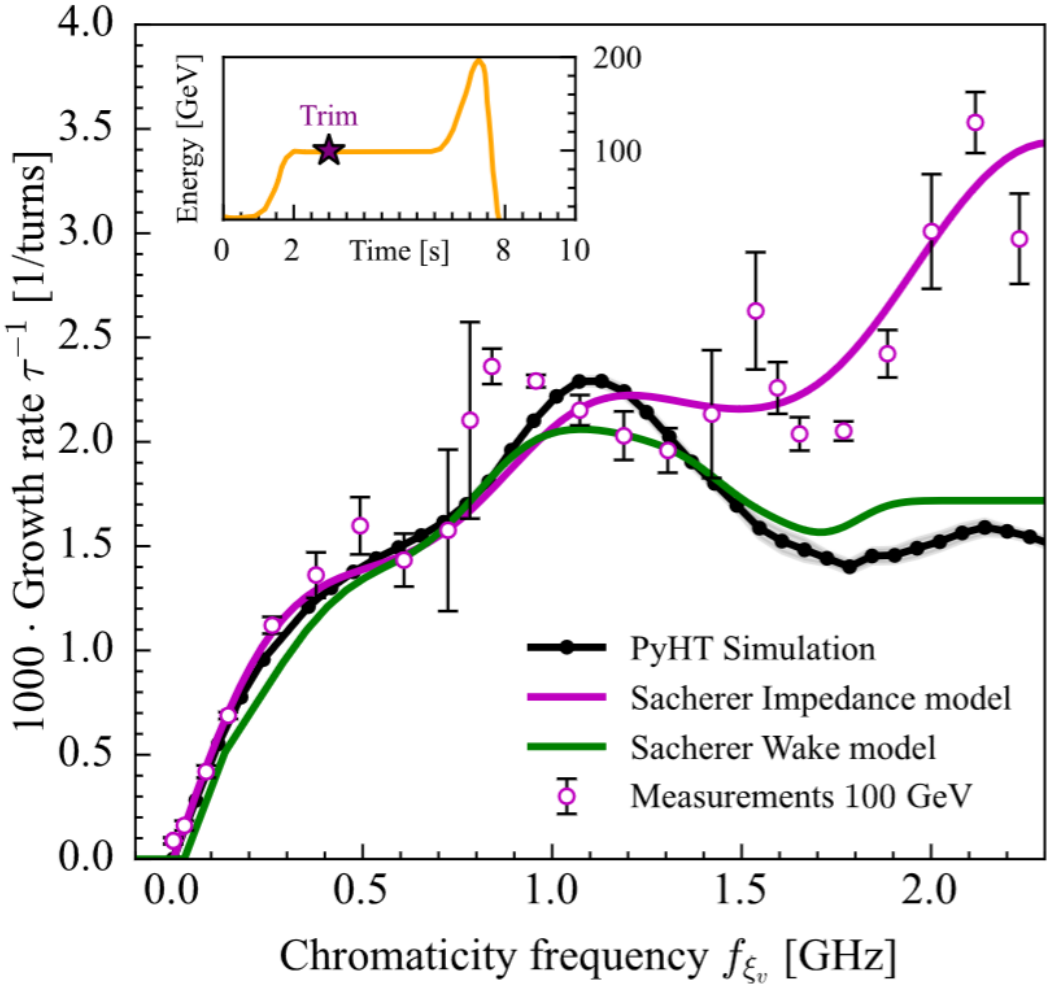
Wakefunction Extrapolation

SPS Transitions



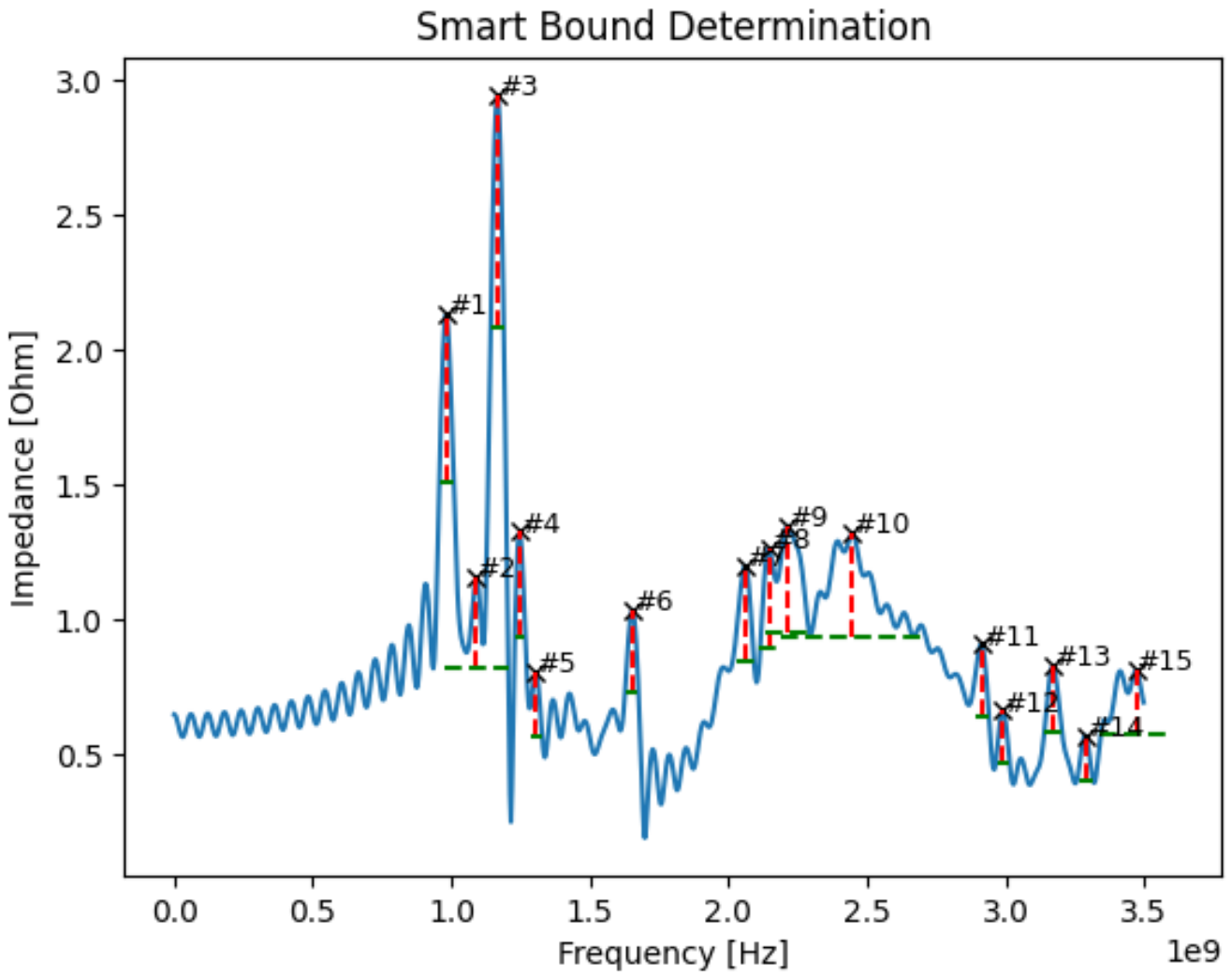
- Wakepotential of nine different types of transitions numerically simulated

Sacherer Formula



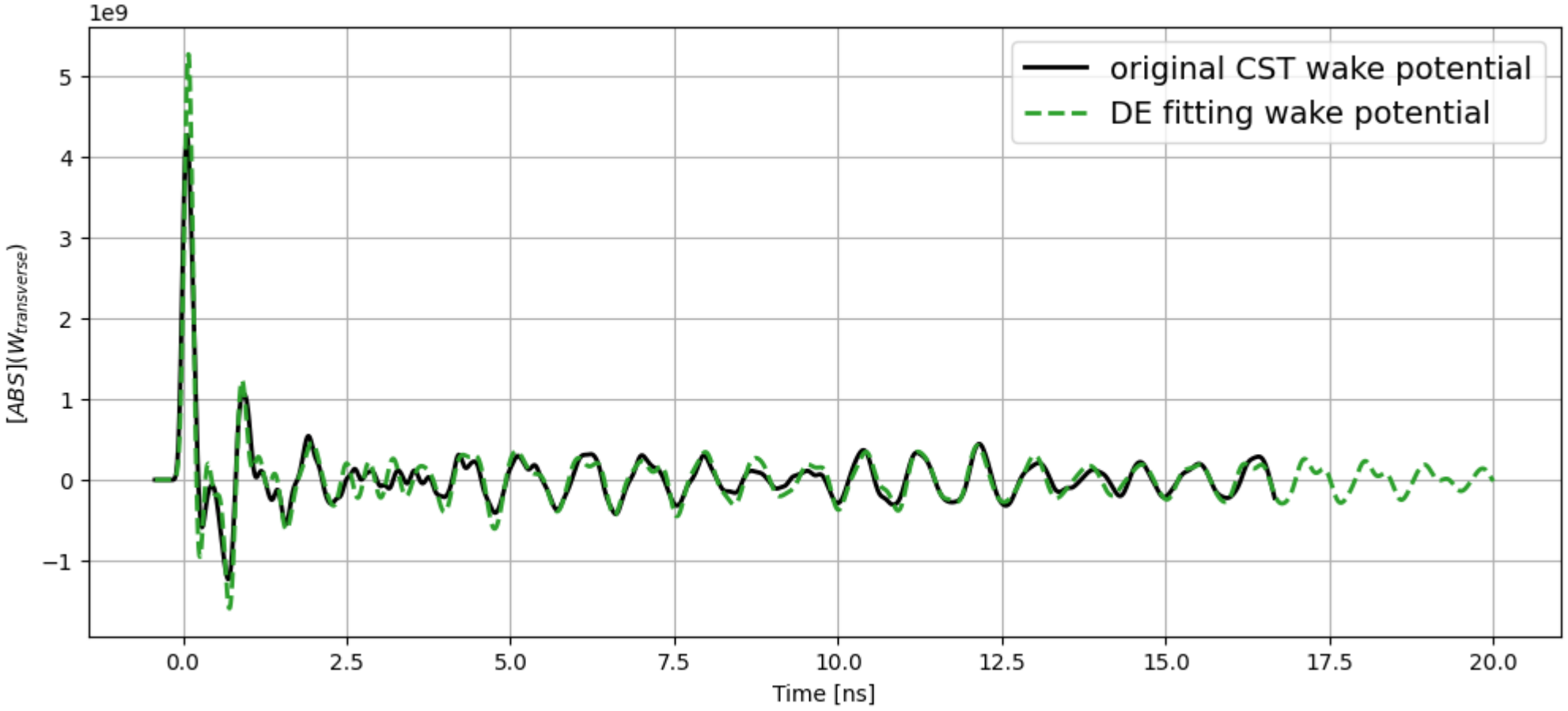
- Wakepotential used as approximation of wakefunction in the SPS wake model
- Wakepotential treated as wakefunction for green curve
- Wakepotential convoluted with bunch for purple curve
- Difference between wakefunction and wakepotential in analytical calculations

Iddefix



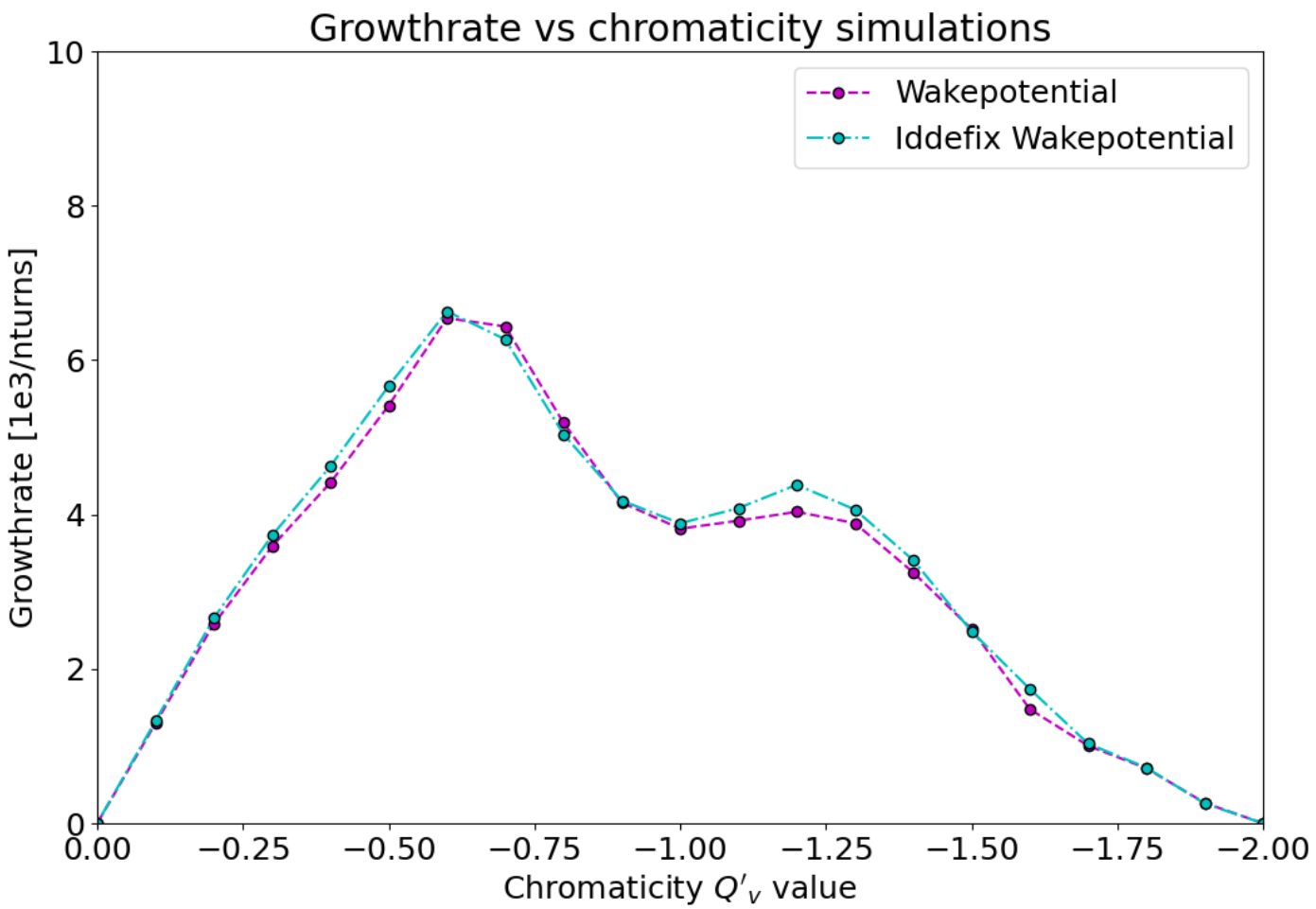
- Used wakepotential for fitting
- Decimated data to 1000 points

Fit with Iddefix



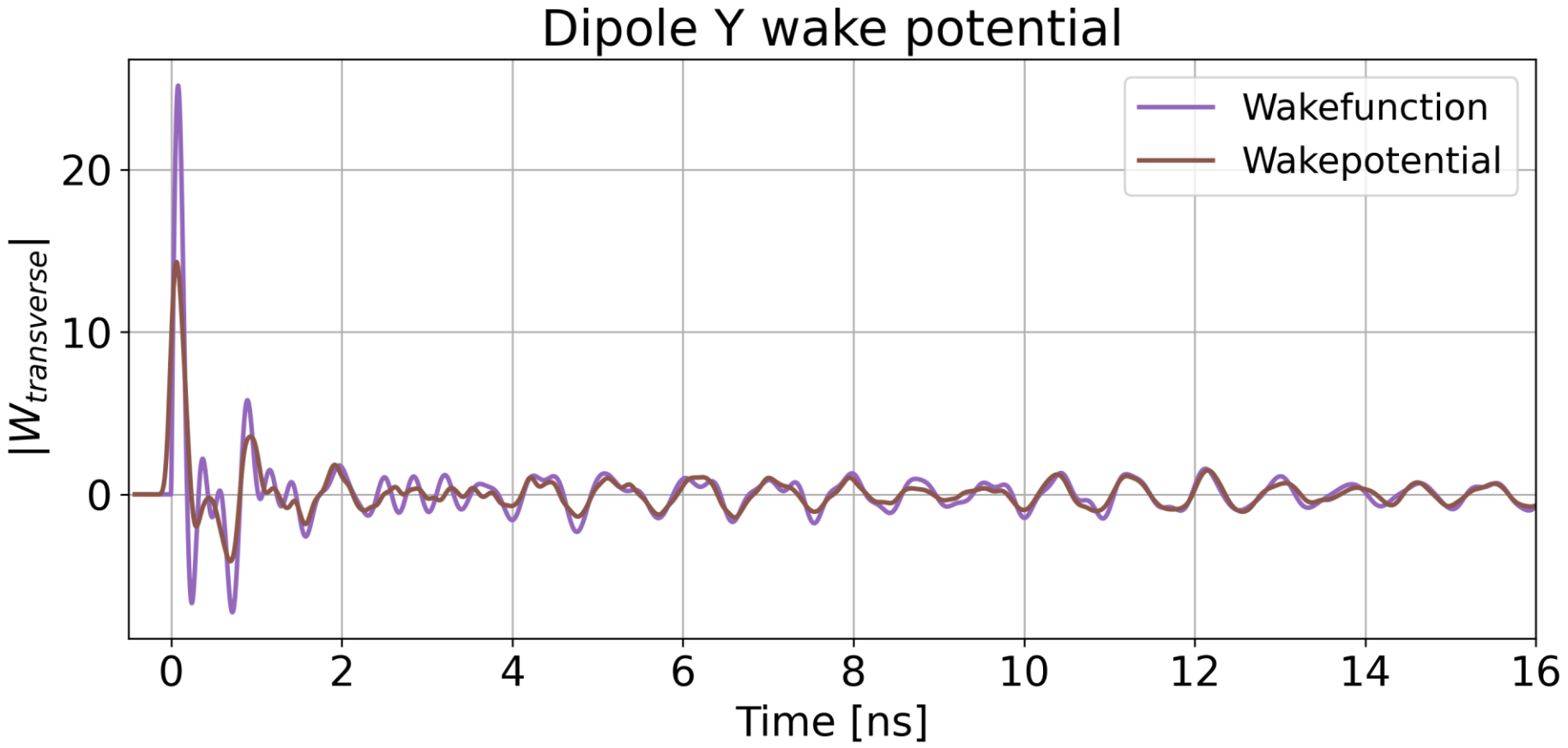
- Good fit of the wakepotential with Iddefix

Validation of Fitted Wakepotential



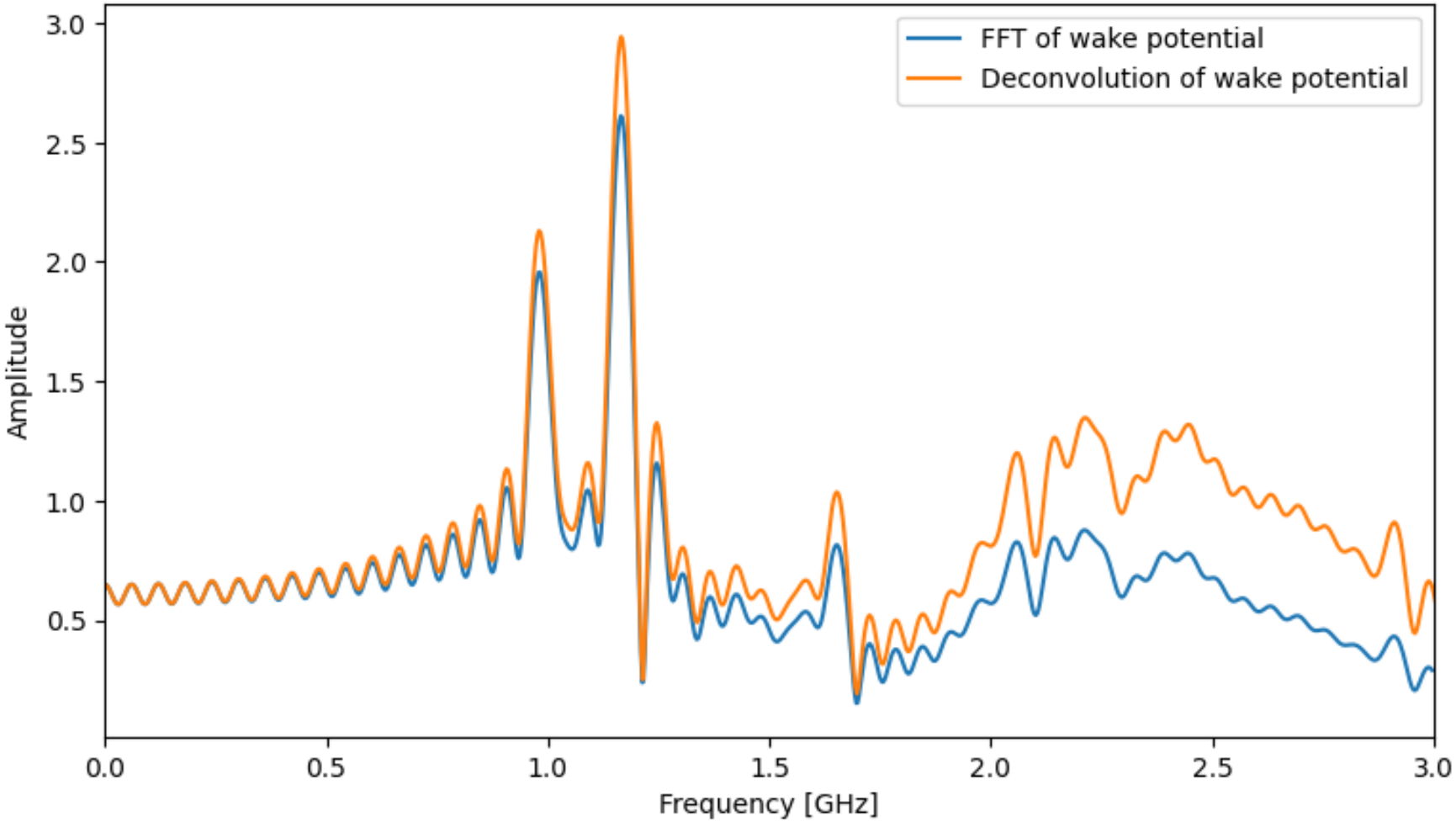
- Wakepotential of transitions replaced by the fitted wakepotential of Iddefix
- Fit of wakepotential sufficient

Extrapolated Wakefunction



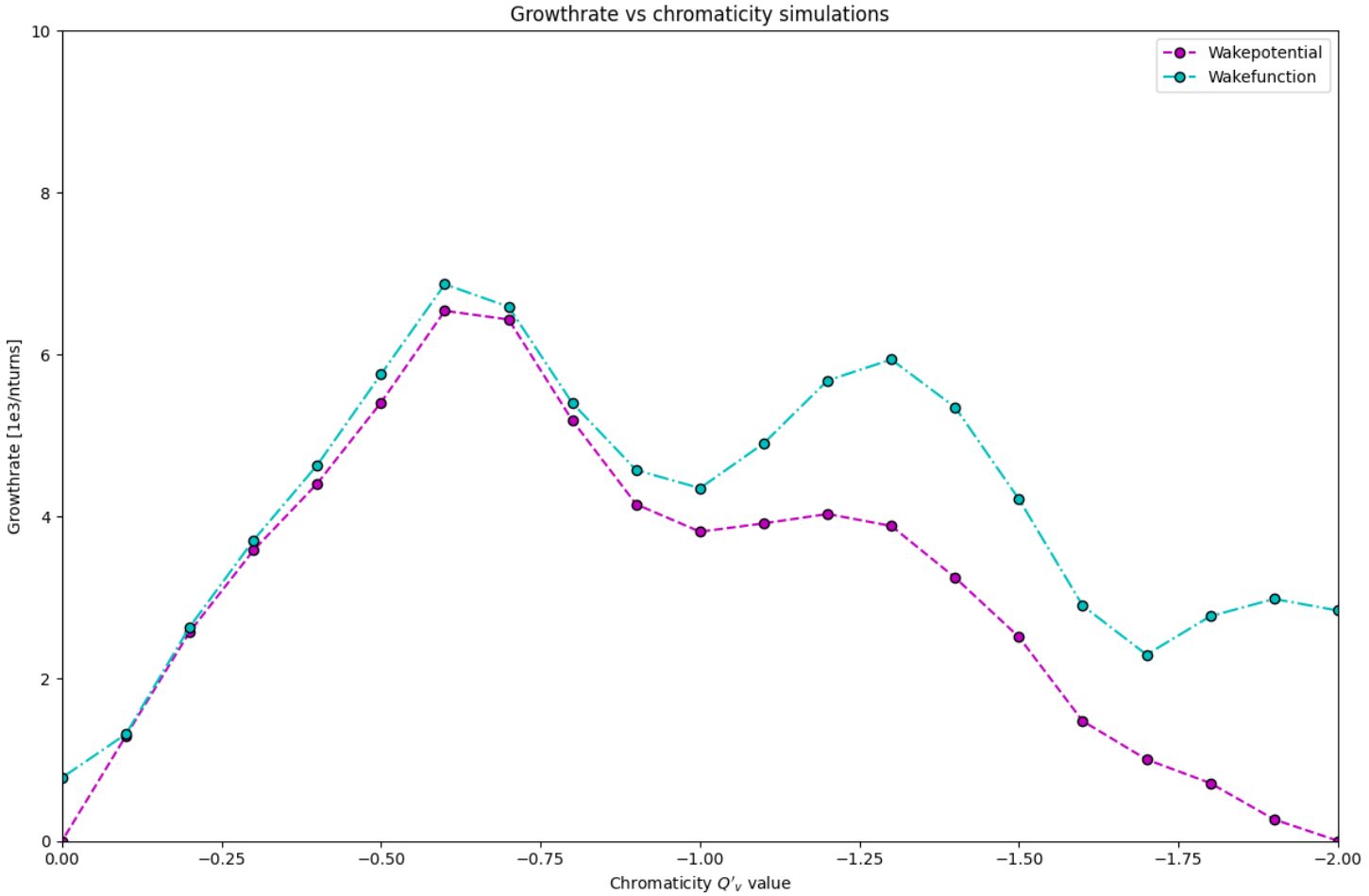
- Wakefunction has higher amplitude and different frequency content

FFT vs. Deconvolution



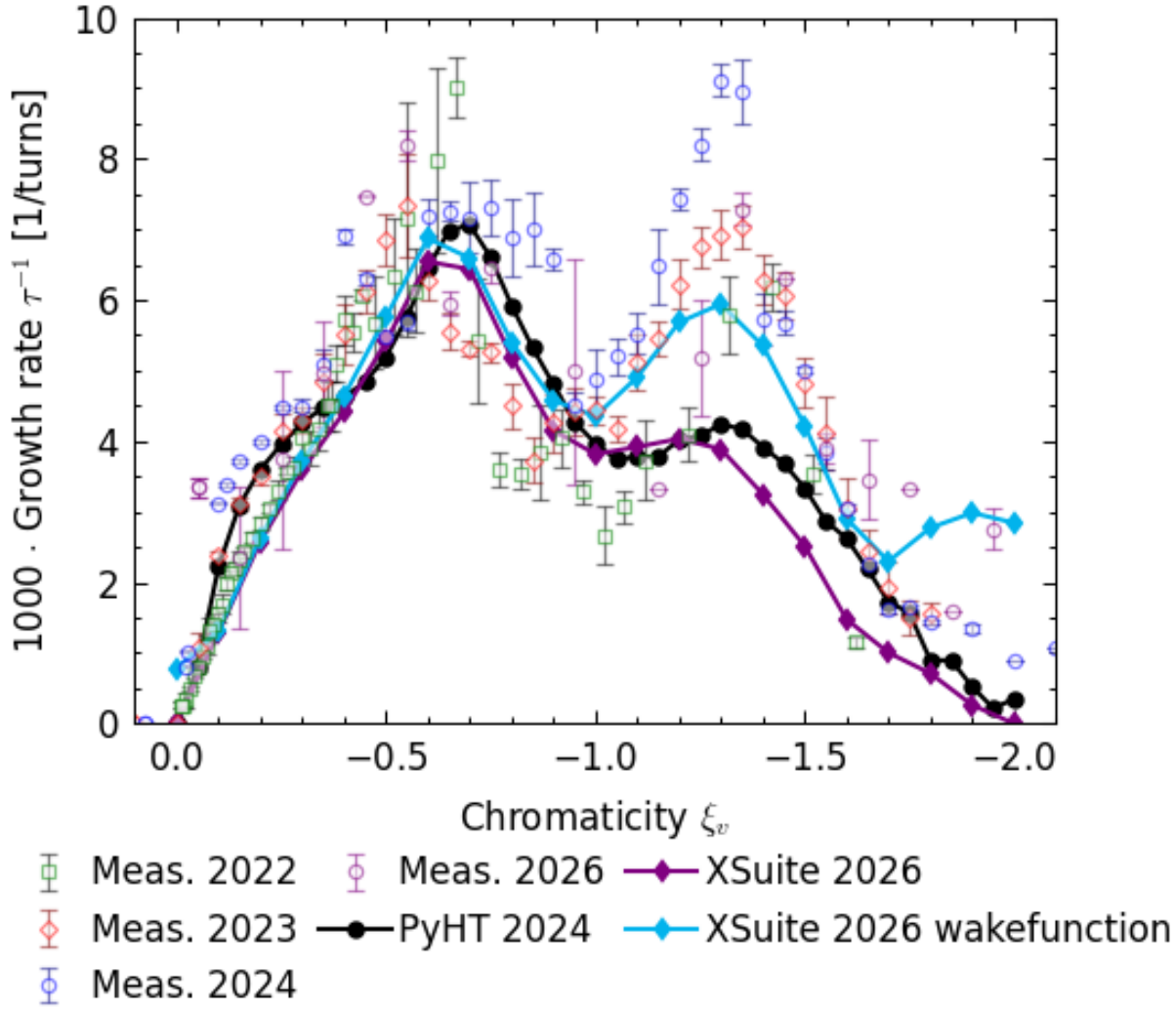
- More high frequency content using the deconvolution

Growthrate Simulation with Wakefunction



- First peak barely changed
- Second peak appears with wakefunction

Simulation with Wakefunction



- Wakefunction shows second peak
- Fits for the measurement 2024
- Overshoots for the other measurements

Conclusion

- **Origin of „Dark Impedance“ found in the approximation of wakefunction of transitions with wakepotential**
- **Second peak appears in simulations with wakefunction**



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